

2016, 2017 and 2019. This may be due to the impact of demonetisation, implementation of GST and the Global pandemic impact of COVID-19 respectively. However, there is not much impact noticed in the case of index options during the same period.

10.6 CURRENCY DERIVATIVE MARKET IN INDIA

Currency derivative is a contract between the seller and buyer, whose value is to be derived from the underlying asset, the currency amount. A derivative based on currency exchange rates is a future contract which stipulates the rate at which a given currency can be exchanged for another currency as at a future date. Currency future contracts allow investors to hedge against foreign exchange risk. Currency Derivatives are available on four currency pairs viz. US Dollars (USD), Euro (EUR), Great Britain Pound (GBP) and Japanese Yen (JPY). Cross Currency Futures and Options contracts on EUR-USD, GBP-USD and USD-JPY are also available for trading in Currency Derivatives segment (NSE).

Foreign exchange derivatives are less active than interest rate derivatives in India, even though they have been around for longer. Over-the-counter (OTC) instruments in currency forwards and swaps are the most popular. Importers, exporters and banks use the rupee forward market to hedge their foreign currency exposure. Turnover and liquidity in this market have been increasing, although trading is mainly in shorter maturity contracts of one year or less (Gambhir and Goel, 2003). In a currency swap, banks and corporations may swap its rupee-denominated debt into another currency (typically the US dollar or Japanese yen), or vice versa. Trading in OTC currency options is still muted. There are no exchange-traded currency derivatives in India.

10.7 LONG-TERM GOVERNMENT BOND AND CORPORATE DEBT MARKET IN INDIA

The corporate bond and long-term Government Securities market is a vital segment of the capital market. The corporate bond market is at the budding phase in India. The corporate bond market remains restricted for participants, largely arbitrage driven and also highly illiquid. The lack of development is anomalous for two reasons: First, India has developed world-class markets for equities and equity derivatives supported by high-quality infrastructure. Second, the infrastructure for the bond market, particularly the government bond market, is similarly of high quality. Until 2007, information on the Indian corporate bond market turnover was incomplete and largely anecdotal. In 2007, however, the Securities and Exchange Board of India (SEBI) launched initiatives to ensure more comprehensive reporting of the over-the-counter (OTC) bond market. Current volumes are running at low levels— around 140 transactions amounting to about USD80 million per day. But corporate bond markets worldwide are typically illiquid, so it may be overly optimistic to expect India to develop a uniquely liquid corporate bond market. Nonetheless, a more liquid market should eventually contribute to lower costs of capital for issuers. India's corporate turnover ratio is quite high at 70 per cent in 2007, comparing favourably with most other emerging East Asian corporate bond markets. However, the small total of outstanding corporate bonds in India means that the secondary market is small and relatively illiquid, irrespective of the turnover ratio (ADB Working Paper Series No. 22).

10.7.1 Outlook for Development of Corporate Debt Market

The development of a corporate bond market in India has lagged in comparison with other financial market segments owing to many structural factors. While primary issuances have been significant, most of these were accounted for by public sector financial institutions and were issued on a private placement basis to institutional investors. The secondary market, therefore, has not developed commensurately and market liquidity has been an issue.

The Government had constituted a High-Level Committee on Corporate Bonds and Securitisation (Patil Committee) to identify the factors inhibiting the development of an active corporate debt market in India and recommend necessary policy actions. The Committee made several recommendations relating to rationalising the primary issuance procedure, facilitating exchange trading, increasing the disclosure and transparency standards and strengthening the clearing and settlement mechanism in the secondary market. The recommendations have been accepted in principle by the Government, RBI and SEBI and are under various stages of implementation.

The two stock exchanges, namely, the Bombay Stock Exchange (BSE) and the National Stock exchange (NSE), as well as the industry body Fixed Income, Money Market and Derivatives Association of India (FIMMDA) have since operationalised respective trade reporting platforms. While all the exchange trades in corporate bonds get captured by concerned exchange's reporting platform, OTC transactions can be reported on any of these platforms. The aggregated trade information across the platforms is being disseminated by FIMMDA on its website. BSE and NSE have also started order-driven trading platforms in July 2007. In practice, however, trading continues to be largely OTC. SEBI has also implemented measures to streamline the activity in corporate bond markets by reducing the shut period in line with that of G-sec, reducing the size of standard lots to Rs. one lakh and standardising the day count convention. Further, to streamline the process of interest and redemption payments, Electronic Clearing Services (ECS), Real-time Gross Settlements (RTGS) or National Electronic Funds Transfer (NEFT) are required to be used by the issuers.

Further progress is anticipated concerning rationalising the primary issuance procedures, which is a critical step for moving away from the predominance of private placements. To reduce the settlement risk and enhance efficiency, the Patil Committee has also proposed setting up of a robust clearing mechanism. The settlement was proposed to be initially on DVP I basis (i.e. trade by trade basis) to address the counterparty settlement risk and gradually migrate to DvP III (net settlement of funds as well as securities) to impart enhanced settlement efficiency. (The delivery versus payment (DVP) modules can be broadly classified into three broad categories viz. DVP I, DVP II and DVP III. Under DVP I, the fund's leg as well as the securities leg are settled simultaneously on a contract-by-contract basis. Under DVP II, while the securities leg is settled on a contract-by-contract basis, the fund's leg is settled for the net amount). Under DVP III, both the funds and the securities legs are settled for the net amounts.)

Patil Committee has recommended two important measures to be initiated by the Government, namely rationalisation of stamp-duty, and the abolition of tax deduction at source, as in the case of government securities. As the corporate debt markets develop and RBI is assured of the availability of efficient price

discovery through significant increases in public issues as well as secondary market trading, and an efficient and safe settlement system, based on DVP III and STP is in place, RBI is committed to permitting market repos in corporate bonds. In the medium term, considering the overall macro-economic situation, the ceiling for foreign investment in both government securities and corporate debt will continue to be calibrated as an instrument of capital account management. In particular, a more liberalised access to foreign investment would be appropriate when, among other things, an efficient and safe settlement system is well entrenched, aggregate consolidated public debt to GDP ratio reaches a reasonable level, say less than 50 per cent, and the corporate debt market acquires depth and liquidity with a significant role for insurance and pension funds in India. In the past, government securities dominated the debt market in India, partly on account of the fiscal dominance and the absence of contractual savings. In the absence of contractual savings only banks tended to deploy their funds in the corporate bond market, mainly through private placement. RBI is hopeful that the recent slow but steady development of insurance sector, mutual funds etc. coupled with the existence of a reliable government securities market and the availability of robust reporting, trading and settlement mechanisms would lead to the rapid development of a vibrant corporate debt market. A framework for the development is already available through the recommendations of the Patil Committee, the implementation of which has already been taken up by the various agencies (Dr. Y. V. Reddy, 2007).

Check Your Progress 3

- 1) State the reasons for trading in the derivatives.

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- 2) Which instruments are traded in the exchange traded markets?

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- 3) What are the various reforms and recommendations made with respect to the Indian corporate debt market?

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10.8 LET US SUM UP

Globalisation and financial liberalisation in India have ushered in a battery of changes in the financial architecture of the economy, as a result of which the resultant gain of the global integration of domestic and foreign financial markets has thrown open new opportunities but at the same time exposed the financial system to significant risks. While the capital market reforms are impressive, there are still areas that present major problems. The long term debt and corporate debt market present the biggest problems. As a result of which, many large Indian companies look to foreign capital markets for longer-term debt and equity. Despite this, the prevailing economic conditions both at domestic and global levels suggests that the Indian stock market will continue to grow even though the USA and European markets have yet to recover from recession effect. The market ecosystem of the corporate debt market in India is yet to evolve in terms of enabling vibrancy and depth. The enactment of the Insolvency and Bankruptcy Code (IBC), 2016, which incorporates relevant resolution frameworks introduced earlier, is expected to strengthen investors' confidence in the corporate bond market. The 2016-17 Union Budget announcement of setting up of Credit Enhancement Fund could leverage the access of the infrastructure companies into the corporate bond market, complemented by the large exposure framework laid down by the Reserve Bank (RBI, 2019).

10.9 TERM-END EXERCISES

- 1) Discuss the role and scope of a well-developed capital market in the growth process of the economy.
- 2) Trace the developmental reforms of the capital market in India and assess the impact of these reforms in the growth of Equity and the foreign exchange market.
- 3) Discuss the current scenario and shortcomings of the corporate debt market in India.
- 4) Discuss the functions of SEBI. What are the significant steps taken by the SEBI to develop the Indian Equity Market?
- 5) Explain the structure, conduct and performance of the Equity Derivatives market in India.
- 6) Explain the structure and performance of the Indian Currency Derivatives market.
- 7) Explain why the Indian Corporate Debt market is the nascent stage. Suggest necessary reforms.

10.10 KEY WORDS

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|---------------------------|---|---|
| Derivatives | : | Derivatives are financial contracts whose value is linked to the value of an underlying asset |
| Equity Derivatives | : | Equity Derivatives are a type of derivative whose value is derived, at least partly, from underlying equity securities including stocks, stock indexes. |

Foreign Institutional Investor (FII)	: A foreign institutional investor (FII) is an investor or investment fund investing in a country outside of the one in which it is registered or headquartered.
Hawala	: An informal, unregulated mode of transaction, that usually occurs so as to smuggle foreign currency to and fro your country.
Hedging	: Hedging is a risk management strategy employed to offset losses in investments by taking an opposite position in a related asset.
Hybrid Securities	: A hybrid security is a single financial security that combines two or more different financial instruments. They generally combine both debt and equity characteristics.
Over-the-counter (OTC)	: Over-the-counter (OTC) is the trading of securities between two counterparties executed outside of formal exchanges and without the supervision of an exchange regulator.
Speculation	: Speculation is the practice of engaging in risky financial transactions in an attempt to profit from short term fluctuations in the market value of a tradable financial instrument.
Stock Market	: A stock market is a marketplace where buyers and sellers meet to trade i.e. buy and sell shares of publicly listed companies.

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10.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 10.1
- 2) See Section 10.1
- 3) See Section 10.2

Check Your Progress 2

- 1) See Section 10.3
- 2) See Section 10.3
- 3) See Section 10.4
- 4) See Section 10.4

Check Your Progress 3

- 1) See Section 10.5
- 2) See Sub-section 10.5.1
- 3) See Section 10.7

UNIT 11 FISCAL POLICY AND FISCAL RESPONSIBILITY AND BUDGET MANAGEMENT (FRBM) ACT

Structure

- 11.0 Objectives
- 11.1 Introduction
- 11.2 Theoretical Analysis: IS-LM Framework
 - 11.2.1 Implications of IS-LM Framework for Fiscal Policy
- 11.3 Concepts of Fiscal Policy
 - 11.3.1 Objectives of Fiscal policy
 - 11.3.2 Implications of Fiscal policy
- 11.4 Fiscal Policy in India
 - 11.4.1 India's Fiscal Policy Architecture
 - 11.4.2 Fiscal Policy until 1990s
 - 11.4.3 Fiscal Policy 1990s Onwards
- 11.5 The FRBM Act
- 11.6 The Global Financial Crises and the Fiscal Policy
- 11.7 Goods and Services Tax (GST)
- 11.8 Let Us Sum Up
- 11.9 Term-end Exercises
- 11.10 Key Words
- 11.11 References
- 11.12 Answers or Hints to Check Your Progress Exercises

11.0 OBJECTIVES

After going through this unit, you will be able to:

- state the concept and components of fiscal policy;
- examine how changes in government spending and taxation affect equilibrium level of income and interest rate in terms of the IS-LM framework;
- discuss the fiscal policy in India since independence;
- recognise the need to adopt the Fiscal Responsibility and Budgetary Management Act;
- discuss the role of fiscal policy to meet the challenges arisen out of global financial crises 2008; and
- appreciate the genesis of GST, its implementation and trend of GST collection.

11.1 INTRODUCTION

During the aftermath of the Great Depression in 1930s, British economist John Maynard Keynes first argued for the role of government to push the economy out of the slump. His book *The General Theory of Employment, Interest and Money* is the foundation of the Keynesian economics. He argued for the role of fiscal and monetary interventions by the government to influence the aggregate demand. According to Keynes, high unemployment in Britain and the United States was the result of a deficiency in the aggregate demand. At the time of the Great Depression, Keynes favoured fiscal policy measures to stimulate the demand. This started a new school of thought in the discipline of economics, the Keynesian economics.

Much of the debate over economic policy in Great Britain at that time focused on the desirability of government spending on public works as a cure for unemployment, what we would now term an expansionary fiscal policy action. Keynes and others argued that such actions would increase output and employment. Such expenditures would act both directly and indirectly because they would increase the income and hence consumer expenditure of those employed on the public works projects, thus generating secondary employment.

Major tools of Fiscal Policy are:

- a) **Government Spending:** Change in government spending (revenue or capital expenditure) that directly impacts the GDP and is often used as a counter-cyclical measure.
- b) **Taxes:** Change in the taxes such as the changes in direct and indirect taxes can influence the behaviour of agents in the economy and therefore impact the GDP.

In this unit we dwell into understanding the functioning of such policy changes. The unit discusses the theoretical framework for understanding the fiscal policy. For this purpose we shall use the IS-LM framework to examine how changes in government spending and taxation effects equilibrium level of income and interest rate in the economy. Thereafter, we shall discuss India's fiscal policy experience since independence. It is further sub-divided into sections which describe the evolution of fiscal policy until the 1990s, following the major economic reforms of the 1991 till the next step of fiscal discipline and consolidation based on the FRBM Act. Lastly, we shall talk about the new fiscal framework, recent developments in the fiscal policy space including implantation of the GST. The unit ends with the discussion of the 15th Finance Commission and the challenges that the fiscal policy space currently faces.

11.2 THEORETICAL ANALYSIS: IS-LM FRAMEWORK

Let us recall that we have studied IS-LM framework in Unit 2 of the course Macro-economic analysis (MEC-002) in general. In order to examine the impacts of the various tools of fiscal policy on aggregate income, savings and interest, it is desirable to have a re-look at the IS-LM framework.

A central notion in the Keynesian model is that an equilibrium level of output requires that total output be equal to the aggregate demand. In our model, this condition for equilibrium can be expressed as:

$$Y = AD \quad (1)$$

where, Y represents total output (GDP) and AD stands for aggregate demand or desired expenditures on output.

Aggregate demand (AD) consists of three components:

- household consumption (C),
- desired business investment demand (I),
- and the government sector's demand for goods and services (G).

$$\text{Thus, in equilibrium we have } Y = AD = C + I + G \quad (2)$$

The simple form of Equation (2) and of the identities discussed later results from overlooking some complexities in the definitions of GDP and national income. To begin with we are dealing with a 'closed' economy, hence exports and imports do not appear in Equation (2). We later extend the simple model to include the components of exports and imports. We also assume that prices are constant, that is, all the variables are real variables and hence the changes in these variables are also in real terms.

In the Indian context, National Accounts Statistics (NAS) provides the break-up of the GDP components. The GDP (or Y in the model) is calculated as:

$$GDP = PFCE + GFCE + GCF + NX$$

Where, PFCE (C in Equation 2) is private final consumption expenditure by households

GFCE (G in Equation 2) is government final consumption expenditure

GCF (I in Equation 2) is gross capital formation

NX is net exports (exports – imports).

Figure 11.1 shows the share of the major components of the GDP in India since 1990-91. It is clear that the private final consumption expenditure by the household constitutes majority of the GDP. We now look at each of these components in details to understand how changes in fiscal policy influence the GDP.

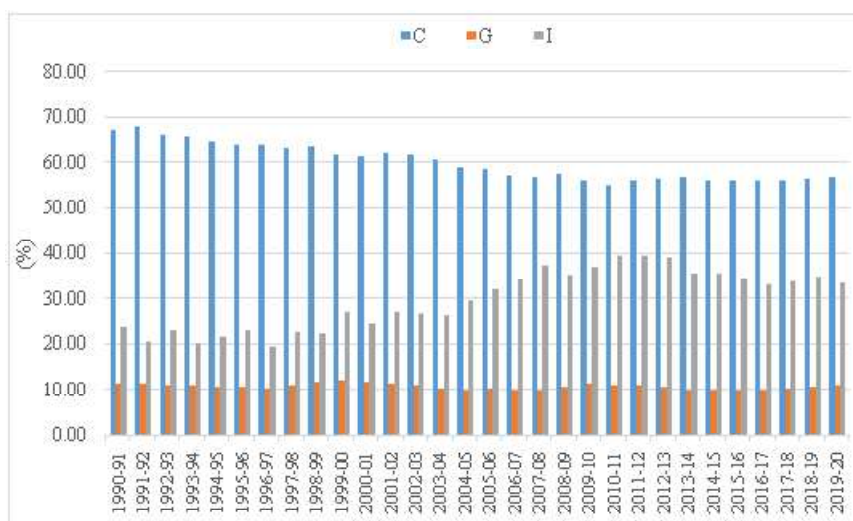


Fig. 11.1: Trends in share of major components of GDP in the overall GDP of India
Source: CMIE

I) Consumption

The private final consumption expenditure (PFCE) by the household constitutes the largest part of the GDP in India, though its share has declined overtime. It formed 56.73 per cent of the total GDP (at constant prices) in 2019-20.

Keynes believed that the level of consumer expenditure is seen as a stable function of disposable income, where disposable income (YD) in our simple model is national income minus net tax payments ($YD = Y - tY$), where t represents the tax rate. The specific form of the consumption-income relationship, termed the consumption function, proposed by Keynes was:

$$C = \bar{C} + c(1-t)Y$$

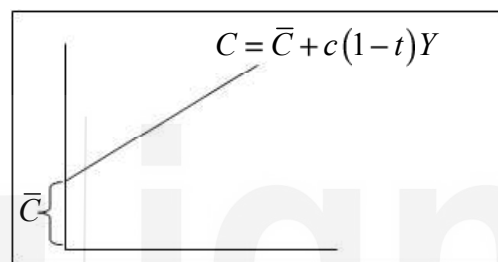


Fig. 11.2: The consumption schedule

The intercept term, which is assumed to be positive, is the value of consumption when disposable income equals zero. As such, it can be thought of as a measure of the effect of variables other than income, i.e. of variables not explicitly included in this simple model on consumption. The parameter c , the slope of the function, gives the increase in consumer expenditure per unit increase in disposable income. In notation, we frequently use

$$c = \Delta C / \Delta Y$$

where, Δ is called delta representing change or difference. It may be recalled that it is the marginal propensity to consume and is a function of several social-cultural and personal factors.

II) Government Spending

Government spending (G) is a second element of aggregate demand. Government spending is assumed to be controlled by the policy maker and therefore does not depend directly on the level of income. It is therefore sometimes represented as $(\bar{G})$ with the bar on top denoting that the variable is constant in the equation. We assume that the level of tax receipts (T) is also controlled by the policy maker and is a policy variable.

III) Investment

Investment is also a key variable in the Keynesian system. Change in the desired business investment expenditure was one of the major factors that Keynes thought was responsible for changes in income. Indeed, investment plays a dual role, it adds to aggregate demand and also to capacity creation, resulting in improved aggregate supply. We take the following function to explain the relationship of investment with the rate of interest (Figure 11.3).